Process: setting observation points

DevSecOps observability tools

Agile project managing dashboards

- Kanban boards
- Burndown charts
- Jira boards
- GitHub project boards.

Application monitoring dashboards

- Jaeger (CNCF): open-source, end-to-end, trace requests, response time, and service mesh.
- Open Census: view data on host but exportable to aggregators.

DevSecOps platform dashboards

- Prometheus: scrape metrics from nodes.
- Grafana: event-driven metrics and multiple panels, correlates multiple inputs.



Source: https://opensource.com/article/18/12/introduction-prometheus



Source: https://opensource.com/article/17/8/linux-grafana

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Observability Push or Pull policies

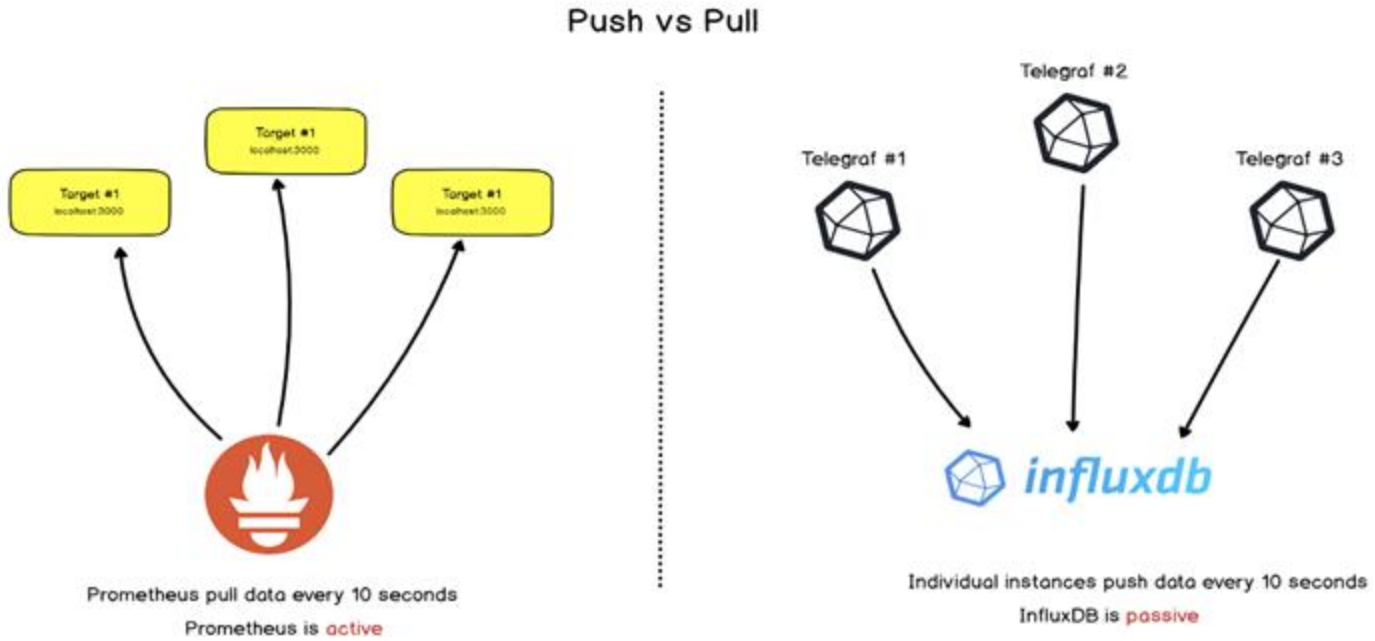
Push to collect

- each app set to push data to central monitor
- can overwhelm storage.

Pull to collect

- active observability
- configure pulling agent
- rate-control variable for different metrics.

- Find a way to ask why it is not working.
- Compile useful insights.
- This is a system-wide behaviour.
- Implement tools like Splunk or ELK ElasticSearch, logstash, Kibana.

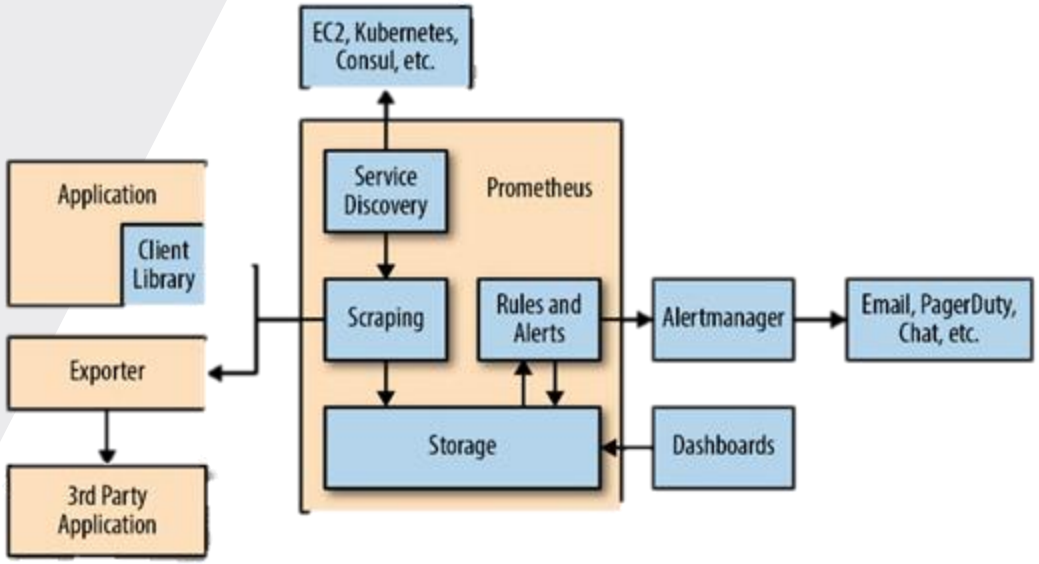


Source: https://devconnected.com/the-definitive-guide-to-prometheus-in-2019/

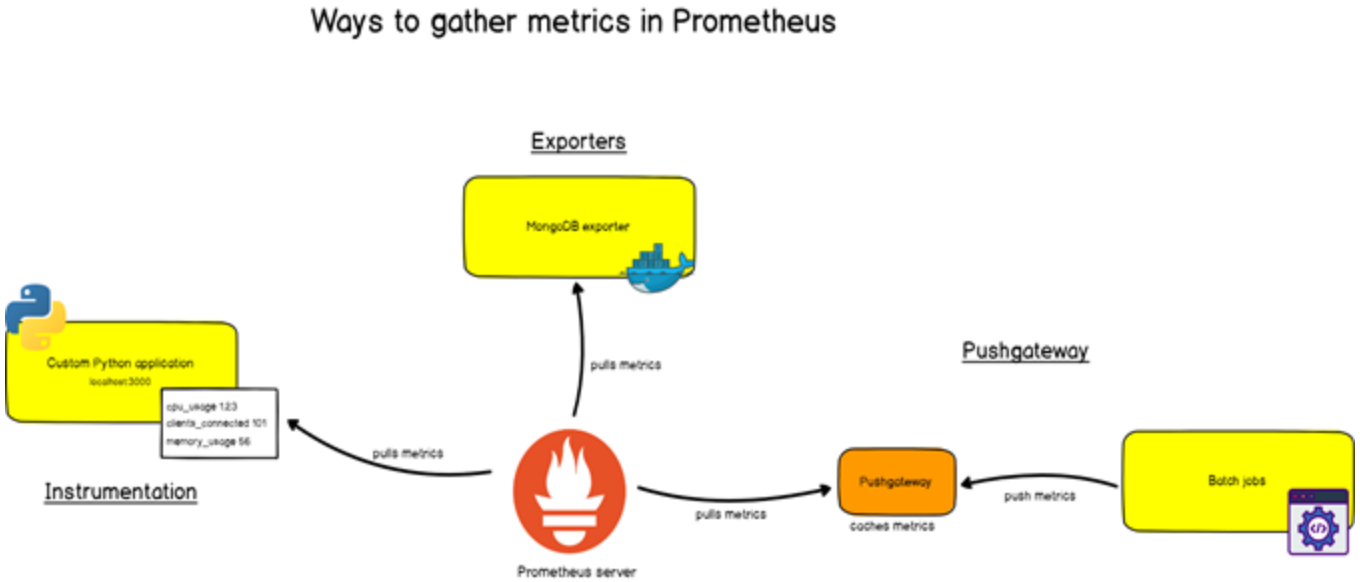
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Security alerts

- Security agents exchange with security monitors metrics data, threshold values, security events.
- Use security event retrospectives to determine, what you want to know about when the event happens?



Source: https://grafana.com/blog/2024/10/11/key-prometheus-concepts-every-grafana-user-should-know/



Source: https://devconnected.com/the-definitive-guide-to-prometheus-in-2019/

- Alerts management process can be manual or automated.
- Example alert designs:
  - dispatch (for example text, email, chat or others.)
  - suppress alerts.
  - thresholds and barriers.
- Do not forget secrets in lookouts.

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Effective monitoring

Employ specifically designed tools for tailored outcomes.

Open policy agent:

- open-source, policy engine with high-level declarative language (Rego)
- sidecar service to verify policies and configuration in:
  - microservices
  - CI/CD pipelines
  - API gateways.
- decouples decision-making from enforcement.

OWASP Tools:

- Zero Attack Proxy: automated vulnerability (ZAP) and supports manual pen test
- PurpleTeam: security regression for pipeline, job file to demonstrate test (uses cloud and local).

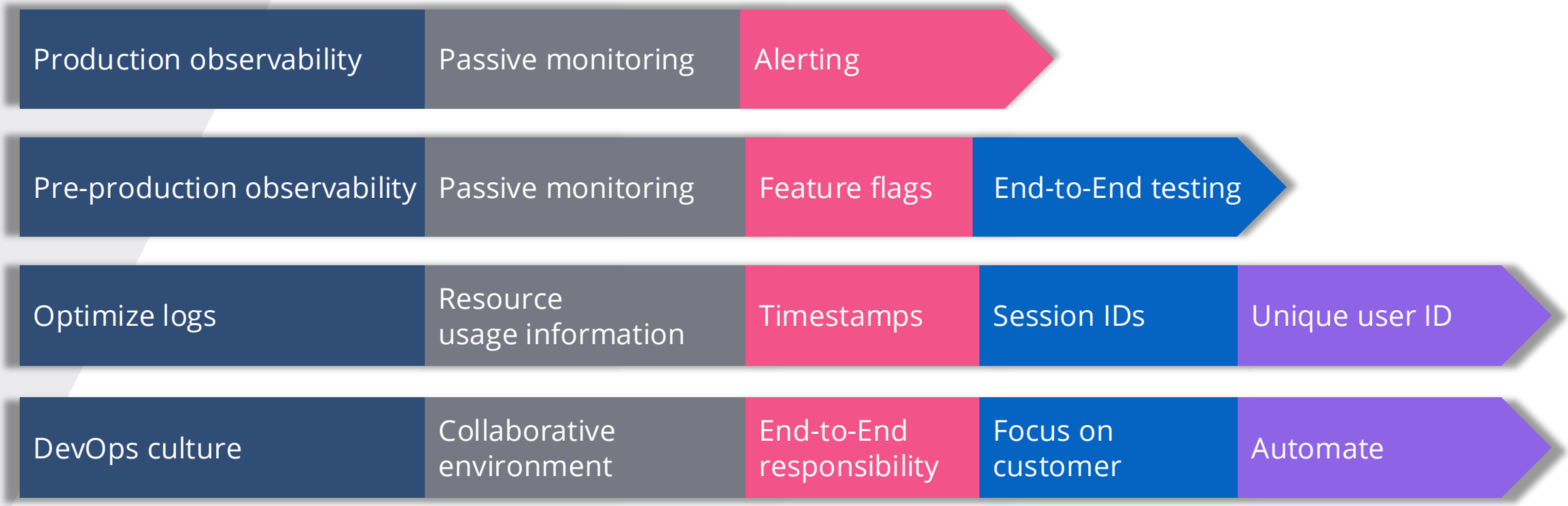
GitHub: Dependabot for dependencies.

Many sites have commercial and open-source offerings.



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Optimizing observability



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